THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 4: **Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a quadratic equation? 1

(a) $x^2 + 2x + 1 = (4 - x)^2 + 3$ **(b)** $x(x + 1) + 8 = (x + 2)(x - 2)$ **(c)** $(x - 2)^2 + 1 = 2x - 3$ **(d)** $x^2 + 3x + 1 = (x - 2)^2$

2. The roots of $x^2 - x - 12 = 0$ are: 1

(a) $4, -3$ **(b)** $-4, 3$ **(c)** $6, -2$ **(d)** $12, -1$

3. The discriminant of $3x^2 - 2x + \frac{1}{3} = 0$ is: 1

(a) 0 **(b)** 4 **(c)** −4 **(d)** 8

4. The nature of the roots of $x^2 + x + 1 = 0$ is: 1

(a) two distinct real roots **(b)** two equal real roots
(c) no real roots **(d)** more than two real roots

5. A student states: "$x^2 - 6x + 9 = 0$ has two distinct real roots." The correct response is: 1

(a) Correct
(b) Incorrect; $D = 0$, so the roots are equal $(3, 3)$
(c) Correct only for $x > 0$
(d) Cannot be determined

6. If $x = 1$ is a root of $kx^2 - 3x + 2 = 0$, then k is: 1

(a) 1 **(b)** −1 **(c)** 2 **(d)** 3

7. The values of k for which $x^2 + kx + 9 = 0$ has equal roots are: 1

(a) ± 3 **(b)** ± 6 **(c)** ± 9 **(d)** 6 only

8. To solve $x^2 - 5x = 0$, Student A divides by x and gets $x = 5$ only; Student B gets $x = 0, 5$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

9. The roots of $ax^2 + bx + c = 0$ $(a \neq 0, b^2 - 4ac \geq 0)$ are given by: 1

(a) $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ **(b)** $\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$ **(c)** $\frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ **(d)** $\frac{-b \pm \sqrt{b^2 - 4ac}}{a}$

10. The product of two consecutive positive integers is 306. The equation representing this is: 1

(a) $x^2 + x - 306 = 0$ **(b)** $x^2 - x - 306 = 0$ **(c)** $x^2 + x + 306 = 0$ **(d)** $2x + 1 = 306$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by factorisation: $2x^2 + x - 6 = 0$. 2

12. Find the nature of the roots of $2x^2 - 6x + 3 = 0$. 2

13. Find the values of k for which $2x^2 + kx + 3 = 0$ has two equal roots. 2

14. A student says that the roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ are $-\sqrt{2}$ and $-\frac{5}{\sqrt{2}}$. Verify. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve using the quadratic formula: $2x^2 - 7x + 3 = 0$. 3

16. Find two numbers whose sum is 27 and product is 182. 3

17. The area of a rectangular field is 528 m². Its length is one metre more than twice its breadth. Find the length and breadth. 3

OR

Find the value of k for which $kx(x - 2) + 6 = 0$ has two equal roots. 3

18. Is it possible to design a rectangular park of perimeter 80 m and area 400 m²? If so, find its length and breadth. 3

OR

Is the following situation possible? “The sum of the ages of two friends is 20 years. Four years ago, the product of their ages was 48.” Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. An express train takes 1 hour less than a passenger train to travel 132 km between Mysore and Bangalore. If the average speed of the express train is 11 km/h more than that of the passenger train, find the average speeds of the two trains and verify. 5

OR

The sum of the reciprocals of Rehman’s ages (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age and verify. 5

20. **Case study:** A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in ₹) was found to be 55 minus the number of toys produced that day. On a particular day the total cost of production was ₹750. 5

(i) Form the quadratic equation.

(ii) Find the number of toys produced that day.

(iii) A student says only 25 toys could have been made. Comment.

OR

Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank. 5

END OF QUESTION PAPER — Total: 40 Marks